

Poly lactic acid

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Poly Lactic Acid (PLA) is a
biodegradable and compostable ...

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What is Polylactid Acid (PLA)

- * Highly versatile thermoplastic polymer
- * Made 100% from renewable resources
- * Lactic Acid is derived from various sources
 - Corn
 - Sugar Beets
 - Wheat

Aliphatic polyester considered biodegradable and compostable (degrading under the action of microorganism in a humid environment to produce biomass and carbon dioxide)

Thermoplastic, high strength polymer which can be made from renewable resources to yield articles as packaging or as biocompatible / bioabsorbable medicals.

AD

Drops of chemistry

Appearance: clear, translucent or opaque pellets; sweet odour

Melting Point: $\approx 140^{\circ}\text{C}$

Water Solubility: approx 20 mg/L at 20°C

n-Octanol Solubility: slight

Yield Sgth (MPa) 70

Elongation at Break (%) 66

Tensile Sgth (MPa) 100-180

Flexural Sgth (MPa) 119

Permeability (gas/cm³ day atm):

O₂, 550

CO₂, 3000

H₂O, 325

Properties

* Insoluble in water, moisture n' grease resistant

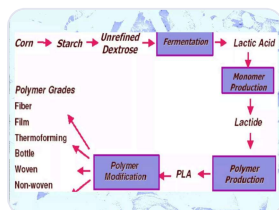
* Biodegradable and compostable

* Clarity and glossiness similar

AD

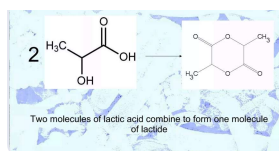
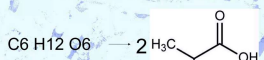
The basic constitutional unit of PLA is Lactic Acid from carbohydrates fermentation or chemical synthesis:

- *Chemical synthesis route is currently used to produce large scale quantities of racemic lactic acid; however, it is economically unviable.
- *Fermentation process can be divided according to the type of bacteria in the process;
 - Heterofermentative; less than 1.8 moles of lactic acid per mole of hexose.
 - Homofermentative; 1.8 moles of lactic acid per mole of hexose. 90+g lactic acid per 100 g glucose.



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Fermentation step

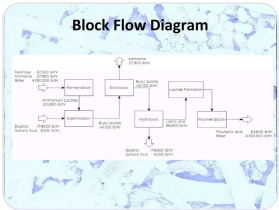


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Polymerization

The lactide polymerizes through ring opening polymerization (ROP) to a molecular weight of approximately 30,000.

But also Direct polycondensation of polylactic acid
- Produces low Mw PLA



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Batch Process: 1-4.5	PH: 5.4-6.4
Continuous Process: 3-9	T: 38-42°C
Cell Recycle Reactors: 76	O2: Avoid due to detrimental effect in the production
Immobilized Cell Reactors: 2.5	Agitation: don't play an important role
Extractive Fermentation: NA	

Continuous reactor

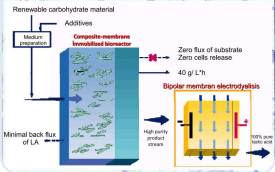
Into the bioreactor at the same time fresh media is added and fluid is removed.

The cells thus continuously propagate on the fresh medium entering the reactor and products, metabolic waste products and cells are removed in the effluent.

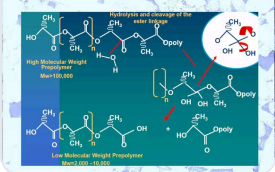
Continuous culture reactors need to be shut down less frequently than batch systems. Cells can also be immobilized in the reactor to maximize their retention and thus increase productivity.

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Extractive fermentation



Degradation



'Unmaking' PLA

- *Fully combustible in composting facilities
- *Can be converted back to monomer
- *Can be completely break down to H₂O, CO₂ and organics
- *Degradation time is weeks or months depending on the conditions

Weeks or Months

AD

Criticisms

- the use of different **additives** in production negate the composting credentials of PLA.
- for medical applications combined with other Bioresins to withstand moisture and higher heat, the **biodegradation rate is slowed** by multiple times.
- made from corn with high **energy waste**, significant **CO₂ release** when manufacturing and during degradation time

Uses and applications

- *Single-use items: plates, cups, film wrap
- *Plastic bottling and fast-food companies
- *Textile industry
- *Paper coatings, Clothing fibers, Compost bags
- *Biomedical field sutures, stents, and dialysis
- *Polylactic acid injections for skin rejuvenation

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Current market

Plastics
2000: 150 million tons
2010: Expected to reach 258 million tons

Biodegradable Plastics
2000: 20 million pounds
2010: Expected to capture 20% of the market for plastics (approximately 50 million tons)

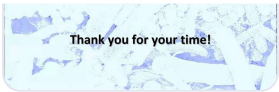
Current selling price of PLA: \$1.50/lb
Current selling price of PET: \$0.60/lb

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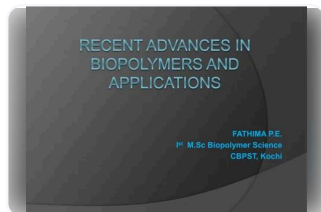
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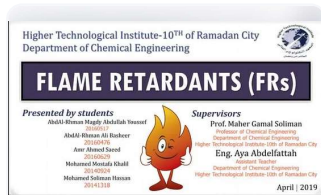
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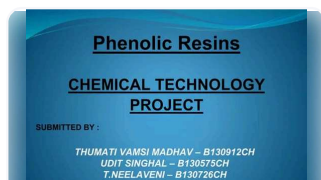
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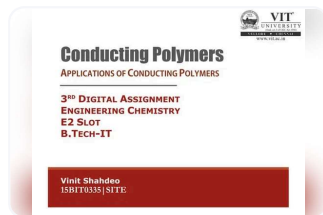


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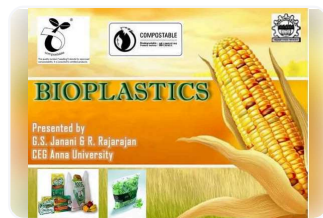
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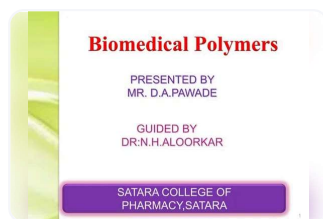
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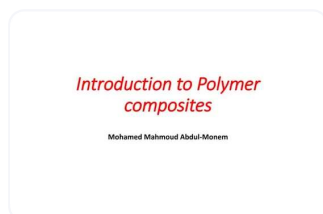
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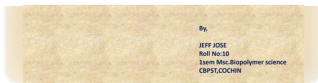
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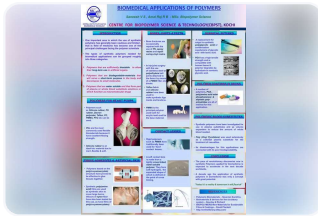
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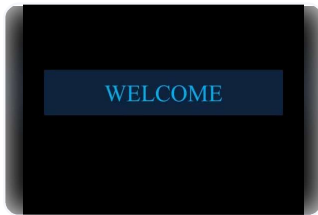
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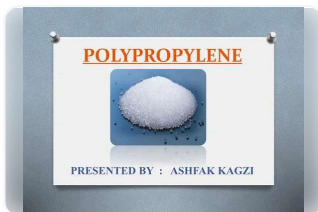
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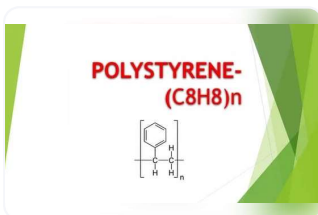
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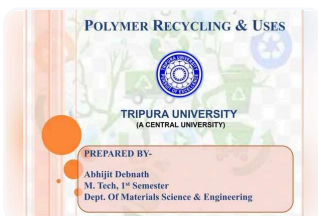
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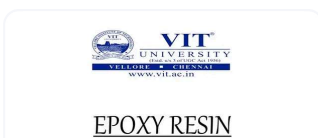


Polystyrene

sai anjaneya

**Polymer Recycling & Uses**

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Epoxy Resin

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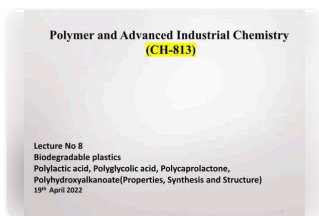
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Polymer nanocomposites

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Bio Plastic

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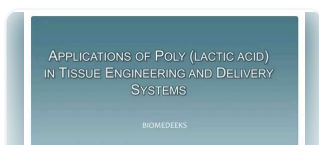
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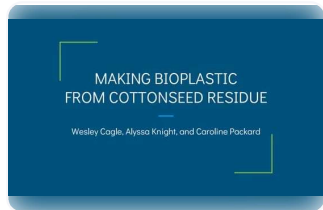
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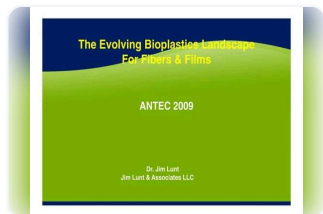
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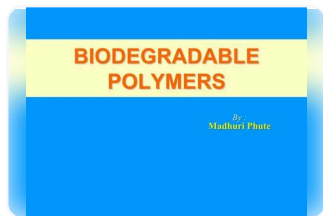
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PLA & amylopectin



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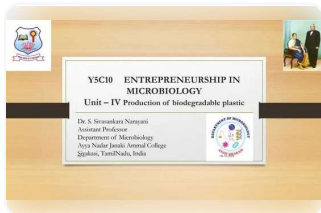


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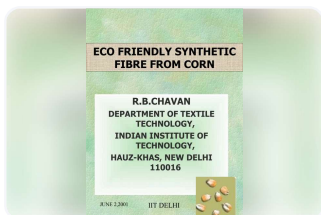
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siva ni



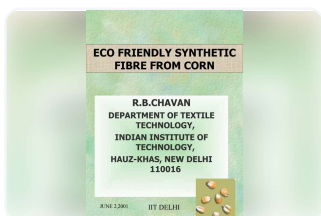
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Marius Murariu



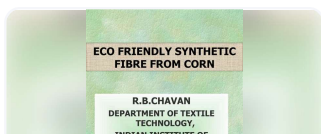
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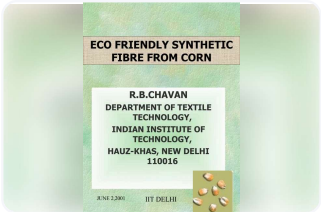
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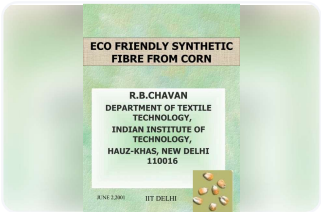
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Abstract

Gioacchino dell'Aquila

1	1.1	1.1.1	1.1.1.1	1.1.1.1.1
2	2.1	2.1.1	2.1.1.1	2.1.1.1.1
3	3.1	3.1.1	3.1.1.1	3.1.1.1.1
4	4.1	4.1.1	4.1.1.1	4.1.1.1.1
5	5.1	5.1.1	5.1.1.1	5.1.1.1.1
6	6.1	6.1.1	6.1.1.1	6.1.1.1.1
7	7.1	7.1.1	7.1.1.1	7.1.1.1.1
8	8.1	8.1.1	8.1.1.1	8.1.1.1.1
9	9.1	9.1.1	9.1.1.1	9.1.1.1.1
10	10.1	10.1.1	10.1.1.1	10.1.1.1.1



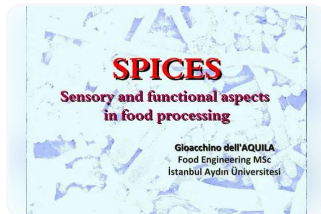
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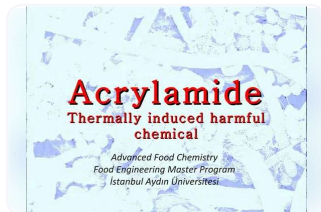
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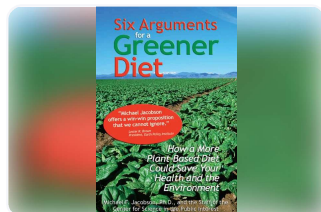
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food spoilage - dstrategies of prevention and new trends for preservation

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From farm to fork - water soluble vitamins

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Poly lactic acid

1. **Poly Lactic** Acid Towards sustainable packaging Gioacchino dell'Aquila Food Engineering MSc İstanbul Aydın Üniversitesi
3. **Background - 1932: Carother** (Dupont) created PLA - 1954: Dupont patented Carothers process - Extremely high cost of manufacturing - 1997: Cargill Dow Polymers LLC forms - 2001: 300 million pound produced at the Blair Nebraska plant
4. **What is Polylactid** Acid (PLA) * Highly versatile thermoplastic polymer * Made 100% from renewable resources * Lactic Acid is derived from various sources - Corn - Sugar Beets - Wheat
5. **Aliphatic polyester considered** biodegradable and compostable (degrading under the action of microorganism in a humid environment to produce biomass and carbon dioxide) Thermoplastic, high strength polymer which can be made from renewable resources to yield articles as packaging or as biocompatible / bioabsorbable medicals.
6. **Drops of chemistry Appearance:** clear, Yield Sgth (MPa) 70 translucent or opaque Elongation at Break (%) 66 pellets; sweet odour Tensile Sgth (MPa) 100-180 Melting Point: < 140°C Flexural Sgth (MPa) 119 Water Solubility: apprx Permeability (mil/m .day.atm): 2 20 mg/L at 20°C O₂, 550 n-Octanol Solubility: slight CO₂, 3000 H₂O, 325
7. **Properties Insoluble in water,** moisture n' grease resistant Biodegradable and compostable Clarity and glossiness similar Requires 20 to 50% less fossil fuels to produce Comparable physical properties to polyethylene terephthalate (PET)
8. **The basic constitutional** unit of PLA is Lactic Acid from carbohydrates fermentation or chemical synthesis: *Chemical synthesis route is currently used to produce large scale quantities of racemic lactic acid; however, it is economically unviable. *Fermentation process can be divided according the type of bacteria in the process; - Heterofermentative; less than 1.8 moles of lactic acid per mole of hexose. - Homofermentative; 1.8 moles of lactic acid per mole of hexose. 90+g lactic acid per 100 g glucose.
10. **Fermentation steen C6 H12 O6** 2 •Bacteria breaks down one molecule of dextrose to form two molecules of lactic acid
14. **Process: (gr/L*h)** Conditions: Batch Process: 1-4.5 PH: 5.4-6.4 Continuous Process: 3 -9 T: 38-42°C Cell Recycle Reactors: 76 O₂: Avoid due to detrimental effect in Immobilized Cell Reactors: 2.5 the production Extractive Fermentation: NA Agitation: don't play an important role

15. **Continuous reactor Into the** bioreactor at the same time fresh media is added and fluid is removed. The cells thus continuously propagate on the fresh medium entering the reactor and products, metabolic waste products and cells are removed in the effluent. Continuous culture reactors need to be shut down less frequently than batch systems. Cells can also be immobilized in the reactor to maximize their retention and thus increase productivity.

16. **Extractive fermentation** Renewable carbohydrate material Additives Medium Composite-membrane preparation Immobilized bioreactor Zero flux of substrate Zero cells release 40 g/ L*h Bipolar membran electrodyalisis High purity Minimal back flux product 100% pure of LA stream lactic acid

17. **Degradation**

18. **'Unmaking' PLA *Fully combustible** in composting facilities *Can be converted back to monomer *Can be completely break down to H₂O, CO₂ and organics *Degradation time is weeks or months depending on the conditions

19. **Weeks or Months**

20. **Criticisms -the use of** different additives in production negate the composting credentials of PLA. -for medical applications combined with other Bioresins to withstand moisture and higher heat, the biodegradation rate is slowed by multiple times. -made from corn with high energy waste, significant CO₂ release when manufacturing and during degradation time

21. **Uses and applications *Single-use** items: plates, cups, film wrap *Plastic bottling and fast-food companies *Textile industry *Paper coatings, Clothing fibers, Compost bags *Biomedical field sutures, stents, and dialysis *Polylactic acid injections for skin rejuvenation

23. **Current market Plastics 2000: 150** million tons 2010: Expected to reach 258 million tons Biodegradable Plastics 2000: 20 million pounds 2010: Expected to capture 20% of the market for plastics (approximately 50 million tons) Current selling price of PLA: \$1.50/lb Current selling price of PET: \$0.60/lb

24. **References PLA 4030D, 4040D, 4041D** Cargill-Dow LLC. (2000). Polylactic acid as a new biodegradable commodity polymer. Auras, R., (2010). Monomers, Polymers and composites from renewable sources, Belgacem, M.N., Gandini, A., (2008). Polylactic Acid Technology. Henton, D.E., et al., (2010)

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